Exercise 9

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Exercise: Trim and Linearization

a) Give a set of states that suffices to fully describe the rigid body motion of an aircraft. Classify the states as either "longitudinal" or "lateral". Explain (by giving an example) why your particular choice of states is not the only possible solution.

Translation
- V Magnitude of Vel — V —" — $(u_K)^O_B$ North Vel
- α Angle of Attack (longitudinal) — γ climb angle / flight path angle — $(v_K)^O_B$ East Velocity
- β Angle of Sideslip (lateral) — χ course angle — $(w_K)^O_B$ Down Velocity

Rotation
- p roll rate (lateral)
- q pitch rate (longitudinal)
- r yaw rate (lateral)

Attitude
- ϕ (lat) Bank Angle — Quaternions q_0, q_1, q_2, q_3
- θ (lon) Pitch Angle
- ψ (lat) Heading Angle / Azimuth

Position
- x_E — λ "Länge"
- y_E — μ
- z_E — h

b) State the primary control effectors of an aircraft. State the primary effect of each control (force or moment)?

Name	Symbol	Primary Effect
Aileron	ξ	Roll Moment
Rudder	ζ	Yaw Moment
Elevator	η	Pitch Moment
thrust lever	δ_T	x_B - Force

c) Transform the well-known nonlinear differential equation of the spring-mass-damper system

$$m \cdot l^2 \cdot \ddot{\varphi}(t) = -m \cdot g \cdot l \cdot \sin\varphi(t) - F_d\left(\dot{\varphi}(t)\right) + F(t) \cdot l \cdot \cos\varphi(t)$$

with the nonlinear damping force $F_d\left(\dot{\varphi}(t)\right) = d \cdot \dot{\varphi}(t) \cdot \cos\frac{\varphi(t)}{\varphi_{max}}$ to a linear system. Linearize around the steady-state operating point $\varphi(0) = \varphi_0$.

Linearization principle:

$\dot{x} = f(x,u)$, $y = h(x,u)$

$$f(x,u) = f(x_0,u_0) + \delta\dot{x} \approx f(x_0,u_0) + \left.\frac{\partial f(x,u)}{\partial x}\right|_{\substack{x=x_0\\u=u_0}} \underbrace{(x - x_0)}_{\delta x} + \left.\frac{\partial f(x,u)}{\partial u}\right|_{\substack{x=x_0\\u=u_0}} \underbrace{(u - u_0)}_{\delta u}$$

$f(x_0,u_0)$: Value at reference point

A: Gradient around the reference point

B

$$\delta\dot{x} = A \cdot \delta x + B \cdot \delta u$$

$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \end{bmatrix} \qquad \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_m} \end{bmatrix}$$

$$\delta\dot{x} = A \cdot \delta x + B \cdot \delta u$$

$$A = \left.\frac{\partial f(\vec{x},\vec{u})}{\partial \vec{x}}\right|_{\substack{\vec{x}=\vec{x}_0 \\ \vec{u}=\vec{u}_0}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}, \quad B = \left.\frac{\partial f(\vec{x},\vec{u})}{\partial u}\right|_{\substack{\vec{x}=\vec{x}_0 \\ \vec{u}=\vec{u}_0}} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \cdots & \frac{\partial f_n}{\partial u_n} \end{bmatrix}$$

$$m \cdot l^2 \cdot \ddot{\varphi}(t) = -m \cdot g \cdot l \cdot \sin\varphi(t) - F_d(\dot{\varphi}(t)) + F_z(t) \cdot l \cdot \cos\varphi(t)$$

1) Classify State or Input

2) Reduce the system order:

$x_1 = \varphi$ $\quad u = F(t) \hat{=}$ Input

$x_2 = \dot{\varphi}$

$\dot{x}_1 = x_2 = \dot{\varphi}$ $\cdot f_1$

$$\Rightarrow \quad m \cdot l^2 \cdot \dot{x}_2(t) = -m \cdot g \cdot l \cdot \sin(x_1(t)) - d \cdot x_2(t) \cdot \cos\left(\frac{x_2(t)}{x_{2,max}}\right) + u(t) \cdot l \cdot \cos(x_1(t))$$

3) Defining the Reference Point / Steady State operating point

$\vec{x}_{ref}$: $\dot{x}_{1,ref} = \dot{x}_{2,ref} = 0$

$x_{1,ref} = \varphi_0$

$x_{2,ref} = \dot{x}_{ref} = 0$

→ obtain our reference input u_{ref} :

$$\dot{x}_{2,ref} = f_2(\vec{x}_{ref}, \dot{\vec{x}}_{ref}, u_{ref}) = -\frac{d \cdot x_{2,ref} \cos\left(\frac{x_{2,ref}}{x_{2,max}}\right)}{m \cdot l^2} - g \cdot \frac{\sin(x_{1,ref})}{l} + u_{ref} \cdot \frac{\cos(x_{1,ref})}{m \cdot l}$$

↳ insert the reference values: $0 = 0 - g \cdot \frac{\sin(\varphi_0)}{l} + u_{ref} \cdot \frac{\cos(\varphi_0)}{ml}$

↳ solve for u_{ref} $\quad u_{ref} = mg \cdot \tan(\varphi_0) = mg \cdot \tan(x_{1,ref})$

4) Derivation of Equation to obtain our A,B Matrices

$$f_1(\vec{x},u) = \dot{x}_1 = x_2$$

$$f_2(\vec{x},\dot{\vec{x}},u) = \dot{x}_2 = -\frac{d x_2 \cos\left(\frac{x_2}{x_{2,max}}\right)}{ml^2} - g \cdot \frac{\sin(x_1)}{l} + u \cdot \frac{\cos(x_1)}{ml}$$

A: $\left.\frac{\partial f_1}{\partial x_1}\right|_{ref}, \left.\frac{\partial f_1}{\partial x_2}\right|_{ref}, \left.\frac{\partial f_2}{\partial x_1}\right|_{ref}, \left.\frac{\partial f_2}{\partial x_2}\right|_{ref}$

$$\left.\frac{\partial f_1}{\partial x_1}\right|_{ref} = 0 \;, \quad \left.\frac{\partial f_1}{\partial x_2}\right|_{ref} = 1 \quad \Rightarrow A = \begin{bmatrix} 0 & 1 \\ ? & ? \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\left.\frac{\partial f_2}{\partial x_1}\right|_{ref} = 0 - \frac{g}{l} \cdot \cos(x_{1,ref}) - \frac{u_{ref}}{ml} \sin(x_{1,ref}) \qquad \left(u_{ref} = mg \cdot \tan(\varphi_0) = mg \cdot \tan(x_{1,ref})\right)$$

$$= -\frac{g}{l}\cos(x_{1,ref}) - \frac{mg \cdot \tan(x_{1,ref})}{ml} \cdot \sin(x_{1,ref})$$

$$= -\frac{g}{l}\cos(x_{1,ref}) - \frac{g}{l}\frac{\sin^2(x_{1,ref})}{\cos(x_{1,ref})} = \frac{-\frac{g}{l}\cos^2(x_{1,ref}) - \frac{g}{l}\sin^2(x_{1,ref})}{\cos(x_{1,ref})}$$

$$= -\frac{g}{l}\frac{1}{\cos(x_{1,ref})}\left(\cos^2(x_{1,ref}) + \sin^2(x_{1,ref})\right) = -\frac{g}{l}\frac{1}{\cos(x_{1,ref})}; \quad \Rightarrow A = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l}\cdot\frac{1}{\cos(x_{1,ref})} & ? \end{bmatrix}$$

$$\left.\frac{\partial f_2}{\partial x_2}\right|_{ref} = -\frac{d \cdot \cos\left(\frac{x_{2,ref}}{x_{2,max}}\right)}{ml^2} + \frac{d \cdot x_{2,ref} \cdot \sin\left(\frac{x_{2,ref}}{x_{2,max}}\right)}{ml^2 \cdot x_{2,max}} \overset{x_{2,ref}=0}{=} -\frac{d}{ml^2};$$

$$\left.\frac{\partial f_2}{\partial x_2}\right|_{ref} = -\frac{d\cdot\cos\left(\frac{x_{2,ref}}{x_{2,max}}\right)}{mL^2} + \frac{d\cdot x_{2,ref}\cdot\sin\left(\frac{x_{2,ref}}{x_{2,max}}\right)}{mL^2\cdot x_{2,max}} = -\frac{d}{mL^2};$$

⇒ Assemble A-Matrix: $A = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L}\cdot\frac{1}{\cos(x_{1,ref})} & -\frac{d}{mL^2} \end{bmatrix}$

B: $\left.\frac{\partial f_1}{\partial u}\right|_{ref} = 0$

$\left.\frac{\partial f_2}{\partial u}\right|_{ref} = \frac{\cos(x_{1,ref})}{mL}$ $\qquad B = \begin{bmatrix} 0 \\ \frac{\cos(x_{1,ref})}{mL} \end{bmatrix}$

⇒ Assemble State Space System:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L}\cdot\frac{1}{\cos(x_{1,ref})} & -\frac{d}{mL^2}\end{bmatrix}\cdot\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ \frac{\cos(x_{1,ref})}{mL}\end{bmatrix}\cdot u$$

$$\dot{\vec{x}} = A\cdot\vec{x} + B\cdot u$$

d) Linearize the equation for the y-position (which is decoupled from the longitudinal motion) and the equation for the bank angle around the condition: „steady-state straight flight without sideslip".

$$0 = \dot{y} - V\cdot\sin\chi\cdot\cos\gamma = \dot{y} - V\cdot\sin(\Psi+\beta)\cdot\cos(\Theta-\alpha)$$

$$0 = \dot{\phi} - p - q\cdot\sin\Phi\cdot\tan\Theta - r\cdot\cos\Phi\cdot\tan\Theta$$

- Lateral Motion States → "decoupled from the longitudinal motion" ⇒ all long. motion states have no influence

→ Subtask a): β, p, r, ϕ, ψ

- "Steady State Straight flight without sideslip" → $\beta_0 = 0$, $p = q = r = 0$, $\phi_0 = 0$

Resist Equations:

$$0 = \dot{y} - V\cdot\sin(\Psi+\beta)\cdot\cos(\Theta-\alpha)$$

$$\hookrightarrow \dot{y} = V\cdot\sin(\Psi+\beta)\cdot\cos(\Theta-\alpha) \hat{=} f$$

$$\frac{\partial f}{\partial p} = \frac{\partial f}{\partial q} = \frac{\partial f}{\partial r} = \frac{\partial f}{\partial u} = 0$$

$$\left.\frac{\partial f}{\partial \beta}\right|_0 = V_0\cdot\cos(\Psi_0+\beta_0)\cos(\Theta_0-\alpha_0) \quad ; \quad \frac{\partial f}{\partial \psi} = V_0\cdot\cos(\Psi_0+\beta_0)\cdot\cos(\Theta_0-\alpha_0)$$

($\beta_0 = 0$)

$\frac{\partial \dot{\beta}}{\partial \beta}\Big|_0$ $\quad \beta_0 = 0$

$$\frac{\partial \dot{\beta}}{\partial \beta}\Big|_0 = V_0 \cdot \cos(\psi_0)\cos(\theta_0 - \alpha_0) \quad ; \quad \frac{\partial \dot{\beta}}{\partial \psi} = V_0 \cdot \cos(\psi_0)\cos(\theta_0 - \alpha_0)$$

$$\delta\dot{\psi} = \begin{bmatrix} \frac{\partial \dot{\beta}}{\partial \beta}\Big|_0 & \frac{\partial \dot{\beta}}{\partial p}\Big|_0 & \frac{\partial \dot{\beta}}{\partial r}\Big|_0 & \frac{\partial \dot{\beta}}{\partial \phi}\Big|_0 & \frac{\partial \dot{\beta}}{\partial \psi}\Big|_0 \end{bmatrix} \cdot \begin{bmatrix} \delta\beta \\ \delta p \\ \delta r \\ \delta\phi \\ \delta\psi \end{bmatrix} = \begin{bmatrix} V_0\cos(\psi_0)\cos(\theta_0-\alpha_0) & 0 & 0 & 0 & V_0\cos(\psi_0)\cos(\theta_0-\alpha_0) \end{bmatrix} \begin{bmatrix} \delta\beta \\ \delta p \\ \delta r \\ \delta\phi \\ \delta\psi \end{bmatrix}$$

1D - "Matrix" → "Vektor"

Bank Angle Equation:

$$0 = \dot{\phi} - p - q\cdot\sin(\phi)\cdot\tan(\theta) - r\cdot\cos(\phi)\cdot\tan(\theta) \qquad \left| \begin{array}{l} \beta_0 = 0 \\ p = q = r = 0 \\ \phi_0 = 0 \end{array} \right.$$

$$\rightarrow f(\vec{x}) = \dot{\phi} = p + q\cdot\sin(\phi)\cdot\tan(\theta) + r\cdot\cos(\phi)\cdot\tan(\theta)$$

$$\frac{\partial f}{\partial \beta}\Big|_0 = \frac{\partial f}{\partial \psi}\Big|_0 = \frac{\partial f}{\partial \alpha}\Big|_0 = 0$$

$$\frac{\partial f}{\partial \phi}\Big|_0 = 0 + q_0 \cdot \cos(\phi_0)\cdot\tan(\theta_0) - r_0 \cdot \sin(\phi_0)\cdot\tan(\theta_0) = 0$$

($q_0 = 0$, $r_0 = 0$)

$$\frac{\partial f}{\partial p}\Big|_0 = 1$$

$$\frac{\partial f}{\partial r}\Big|_0 = \cos(\phi_0)\cdot\tan(\theta_0) = \tan(\theta_0)$$

($\phi_0 = 0$)

⇒ Assemble: $\quad \delta\dot{\phi} = 1\cdot\delta p + \tan(\theta_0)\cdot\delta r$